

Molecular Catalyst Synthesis Strategies to Prepare Atomically Dispersed Fe-N-C Heterogeneous Catalysts

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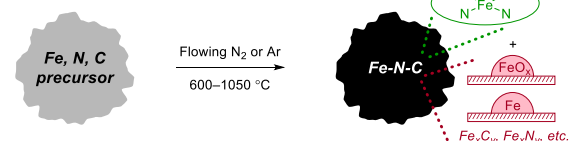
Supporting Information Placeholder

ABSTRACT: We report a strategy to integrate atomically dispersed iron within a heterogeneous nitrogen-doped carbon (N-C) support, inspired by routes for metalation of molecular macrocyclic iron complexes. The N-C support, derived from pyrolysis of a ZIF-8 metal-organic framework, is metalated via solution-phase reaction with FeCl₂ and tributyl amine, as a Brønsted base, at 150 °C. Fe active sites are characterized by ⁵⁷Fe Mössbauer spectroscopy and aberration-corrected scanning transmission electron microscopy. The site density can be increased by selective removal of Zn²⁺ ions from the N-C support prior to metalation, resembling the transmetalation strategy commonly employed for the preparation of molecular Fe-macrocycles. The utility of this approach is validated by the higher catalytic rates (per total Fe) of these materials relative to established Fe-N-C catalysts, benchmarked using an aerobic oxidation reaction.

Several domains of heterogeneous catalysis have benefitted from concepts originating from the field of homogeneous catalysis. Examples include surface organometallic chemistry (SOMC),^{1–4} metal-organic frameworks (MOFs),^{5–8} and various classes of ‘single-atom’ catalysts.^{9–14} These heterogeneous catalysts have active-site structures that resemble molecular catalysts, even while synthetic routes to these materials vary. In many cases, molecular precursors are used to prepare solid catalysts that retain a well-defined molecular fragment within the final structure (e.g., SOMC, MOFs). In other cases, more conventional synthetic strategies (e.g., wet impregnation) are used to distribute isolated active sites onto heterogeneous supports. Atomically dispersed metals incorporated into nitrogen-doped carbon supports (M-N-Cs) are typically prepared by high-temperature (600–1050 °C) pyrolysis of a precursor containing a source of iron, nitrogen, and carbon (Figure 1A). Prominent examples include Fe-N-C catalysts, which show exceptional reactivity as electrocatalysts for the oxygen reduction reaction (ORR)^{13–16} and, more recently, as aerobic oxidation catalysts.^{17–19} The synthetic routes to these materials form porous, graphitic N-doped carbon structures with mononuclear Fe sites, together with other Fe species, such as FeO_x and metallic Fe nanoparticles, among others.²⁰ Recent efforts have identified new multi-step routes to Fe-N-C materials with improved Fe dispersion and site density.^{21–24} These advances are complemented by studies implicating FeN₄ or related mononuclear FeN_x as the active sites for ORR,^{25–27} however, the active sites in aerobic oxidation reactions are less well defined.¹⁹ The long-recognized analogy between the FeN_x sites and the tetragonal N₄ metal binding site in Fe-porphyrins and related Fe-macrocycles^{25,28,29} has prompted consideration of metalation strategies commonly used for the preparation of

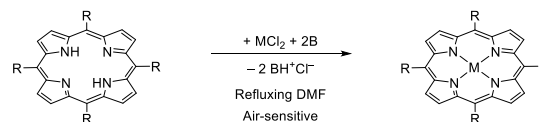
macrocyclic metal complexes that could be adapted to prepare Fe-N-C materials. Molecular precedents highlight two

A. One-step pyrolysis: heterogeneous sites

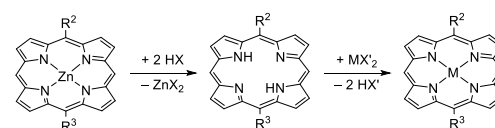


B. Homogeneous metalation of N₄-macrocycles

i. Metalation of free-base porphyrin



ii. Demetalation-metalation of Zn-porphyrin



C. Selective metalation of N_x sites in N-C (this work)

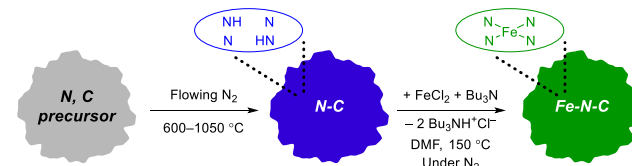


Figure 1. Development of a molecularly inspired Fe-N-C synthesis. (A) Challenges associated with conventional one-step pyrolysis methods. (B) Molecular methods for metalation of homogeneous macrocycles, and (C) extension to Fe-N-C.

complementary strategies (Figure 1B): (i) direct metalation of free-base macrocycles with a metal chloride and a Brønsted base,³⁰ and (ii) stepwise transmetalation of a Zn^{2+} ion, which is commonly used as a template in the synthesis of more complex macrocycles.^{31–33} Feller and coworkers have explored this concept, introducing Fe sites into Mg- and Zn-templated N-C materials.^{22, 34, 35} Here, we identify two protocols directly inspired by the macrocycle metalation conditions, using a Brønsted base to deprotonate N_x surface sites and facilitate Fe metalation of N-C materials (Figure 1C). The resulting Fe-N-C heterogeneous catalysts show a high degree of Fe dispersion that correlates with improved catalytic aerobic oxidation reactivity.

A ZIF-8 MOF was synthesized by reported methods³⁶ and pyrolyzed under flowing N_2 at 1050 °C for 1 h to generate an N-C support, denoted ZIF-8-C (see section 2, Supporting Information for details). The ZIF-8-C retains its particle morphology after pyrolysis (see Figure S7), as observed previously,³⁷ and retains a high degree of microporosity (see Figure S14). ZIF-8-C was then subjected to solution-phase metalation conditions by adapting a protocol reported previously for metalation of free-base porphyrins³⁰ (Figure 2A). Specifically, FeCl_2 was mixed with an organic Brønsted base (0.2 M Bu_3N) in DMF with the ZIF-8-C, heated to 150 °C, and stirred under N_2 overnight. The resulting solid was recovered by filtration, washed with DMF, acetic acid, and methanol to remove soluble and weakly-bound Fe species, and then dried under vacuum. Exclusion of air proved to be important to minimize the deposition of FeO_x clusters that are difficult to remove from the support by solvent rinsing, which is a less important consideration for homogeneous macrocycles. The material synthesized via metalation is referred to as 0.3Fe-ZIF-8-C[FeCl_2], where the leading 0.3 denotes the bulk Fe content (wt%) measured by ICP-OES analysis, and the [FeCl_2] suffix denotes the metalation precursor and conditions used (Figure 2A). This material will be abbreviated as “Fe-N-C^{dm}” (^{dm} = “direct metalation”) hereafter. An alternative method has been

reported using $\text{Fe}(\text{OAc})_2$, relying on the basicity of the acetate anion rather than including exogenous base.³⁸ This protocol was also explored but led to a lower Fe loading (see section 2.4, Supporting Information, for details).

The structure of Fe in Fe-N-C^{dm} after metalation was analyzed by ^{57}Fe Mössbauer spectroscopy and aberration-corrected scanning transmission electron microscopy (AC-STEM). The ^{57}Fe Mössbauer spectrum (Figure 2B) was collected at low temperature (10 K) in order to detect the sextet magnetic splitting that arises from small FeO_x aggregates (<10 nm),^{39–42} differentiating them from atomically dispersed Fe species (see Figure S3, Supporting Information). In addition to the FeO_x sextet feature, denoted “S1”, three additional doublet features were evident in the ^{57}Fe Mössbauer spectrum. Similar species have been identified previously in Fe-N-C materials,^{21,27,39,43–45} and they have been assigned “D1,” an $\text{Fe}^{\text{III}}\text{N}_x$ species^{27,45}; “D2,” an $\text{Fe}^{\text{II}}\text{N}_x$ species^{27,45}; and “D3,” an FeCl_2 species²¹ (see section 3.2 of the Supporting Information for additional information on fitting parameters). AC-STEM images revealed the presence of nanoparticles that exhibit the ordered lattice of crystalline aggregates and an elemental composition consistent with FeO_x (Figure 2C; see Figure S6, Supporting Information for elemental mapping with energy dispersive X-ray spectroscopy (EDS)). Atomically dispersed metal species (Fe or Zn) were also detected in the thin regions at the edge of N-doped carbon support particles as bright dots (Figures 2D and S7).

The ^{57}Fe Mössbauer spectrum is interpreted semi-quantitatively to indicate that approximately 50% of the Fe in Fe-N-C^{dm} is present as atomically dispersed species while the remainder is deposited as FeO_x aggregates (see section 3.2 of the Supporting Information for a discussion of assumptions). The FeO_x and FeN_x species detected by ^{57}Fe Mössbauer are corroborated by the AC-STEM images in Figure 2C and 2D. Taken together, the data indicate that the low-temperature, solution-phase metalation conditions directly generate FeN_x sites, without requiring a subsequent high-temperature pyrolysis step.^{22,23}

Some portion of the Zn originating from the MOF precursor remains in this material (1.2 wt% Zn, Table S2, Supporting Information) and represents a fraction of the atomically dispersed species observed by AC-STEM (Figure 2D). These putative ZnN_x sites²¹ prevent access to N_x sites by FeCl_2 and formation of additional FeN_x sites under the low-temperature metalation conditions. Previous reports have demonstrated replacement of Fe by Zn during high-temperature heat treatment (750–1000 °C)^{21–23} The molecular precedents noted in Figure 1B-ii raised the possibility of an alternative transmetalation strategy, in which acid-promoted removal of the Zn ions could be followed by low-temperature metalation.

Preliminary efforts revealed that solution-phase treatment of Zn-N-C materials with HCl does not reduce the content of Zn ions below 1 wt%. On the other hand, ZnCl_2 has a boiling point of 732 °C, and this property was exploited to promote (partial) removal of Zn ions by flowing gas-phase HCl, carried by an N_2 stream, at 750 °C (Figure 3A, first step). This approach was tested with various precursors, and the most effective used a Zn-N-C material obtained from pyrolysis of ZIF-8 impregnated with 1,10-phenanthroline (Phen) to support additional ligation of Zn. After initial pyrolysis of ZIF-8-Phen at 1050 °C, gas-phase HCl treatment of the resulting ZIF-8-Phen-C material decreased the bulk Zn content in the material from 1.5 wt% to

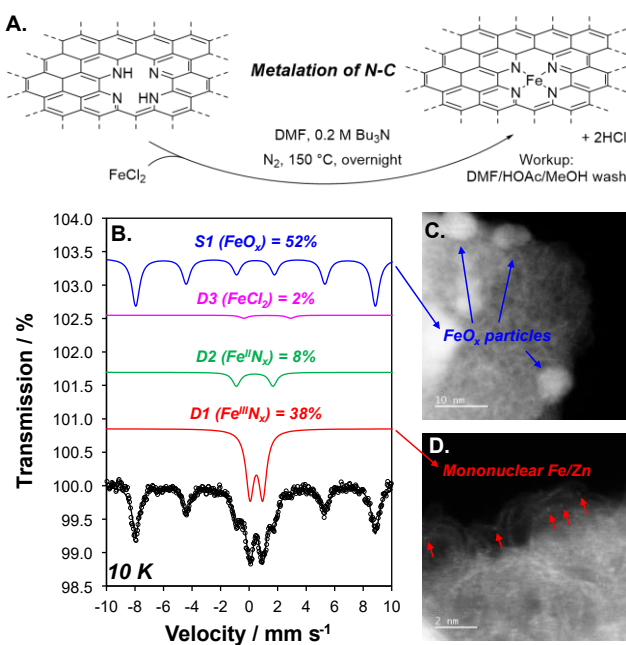


Figure 2. Development of molecularly inspired metalation synthesis of Fe-N-C. (A) Metalation of ZIF-8-C with FeCl_2 . (B) ^{57}Fe Mössbauer spectroscopy (10 K) and (C, D) AC-STEM images of Fe-N-C^{dm}.

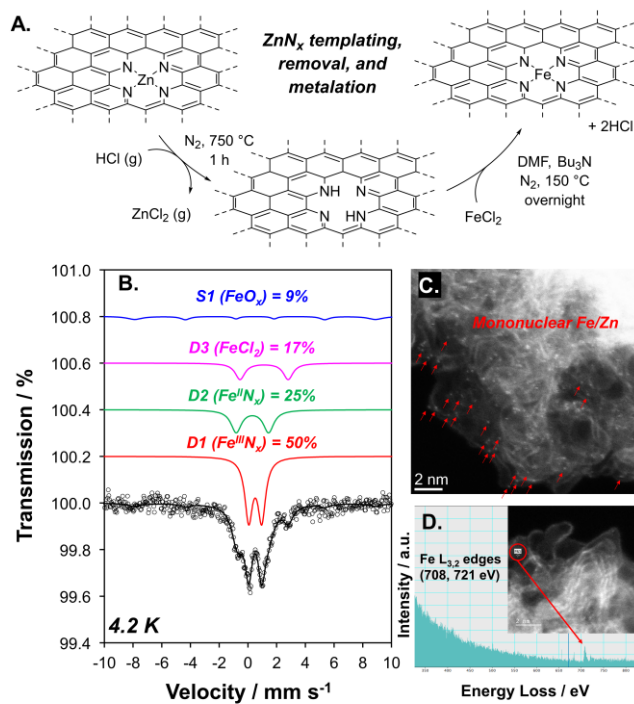


Figure 3. Achieving higher MN_x densities by Zn templating and removal. (A) Proposed chemical steps of Zn removal via gas-phase HCl treatment. (B) ⁵⁷Fe Mössbauer spectrum (4.2 K), (C) AC-STEM images, and (D) EELS analysis of Fe-N-Ctm.

0.9 wt% (see section 2.3 and Table S2 in the Supporting Information for additional details). Some ZnN_x centers are evident in AC-STEM images of this material (Figure S9, Supporting Information). The ZIF-8-Phen-C-HCl sample was then subjected to the same low-temperature solution-phase metalation conditions with FeCl₂ (Figure 3A, second step), affording an Fe-N-C material with 0.3 wt% Fe, on the basis of ICP-OES analysis. This 0.3ZIF-8-Phen-C-HCl[FeCl₂] material is abbreviated as Fe-N-Ctm (tm = “transmetalation”) hereafter.

Fe-N-Ctm was then characterized by ⁵⁷Fe Mössbauer spectroscopy and AC-STEM. The 4.2 K Mössbauer spectrum reveals the same peaks associated with mononuclear FeN_x species (D1, D2) and FeCl₂ (D3), with a rather weak sextet signal for FeO_x aggregates (Figure 3B; cf. Figure 2B). AC-STEM imaging reveals abundant mononuclear metal species (see Figure S10 for additional images). Electron energy loss spectroscopy (EELS) analysis of one atomically dispersed species confirmed its identity as Fe (Figure 3D).⁴⁶ Acquisition of N 1s X-ray photoelectron spectra (Figure S12 and Table S8) support an increase in the relative abundance of M–N_x centers after metalation.⁴⁷

The transmetalation-based synthesis route leads to an increase in the fraction of FeN_x D1/D2 sites in the Fe-N-C material, relative to the direct metalation route (46% to 75%, respectively; cf. Figures 2B and 3B). The impact of this difference on catalytic performance was evaluated by using an aerobic oxidation reaction as a kinetic benchmark. This reaction involves oxidation of a sulfonated hydroquinone (HQ) to the corresponding quinone, which has been used in a mediated fuel cell⁴⁸ and mechanistically characterized.^{49, 50} Previously reported Fe-N-C materials that have been prepared via conventional pyrolysis methods were also evaluated for comparison. These materials include Fe-Phen-C,⁵¹ Fe-PANI-

C,⁵² Fe-ZIF-8-C,³⁷ and a commercially available Fe-N-C, “Pajarito Powder”.⁵³ Initial rates of HQ oxidation were measured under the conditions specified in Figure 4A and correlated with the Fe content of each catalyst (Figure 4B). Conventional Fe-N-C catalysts oxidize HQ at rates that vary by a factor of five ($1.7\text{--}8.2 \times 10^{-5} \text{ mol g}^{-1} \text{ s}^{-1}$, Figure 4B, red data points). The Fe-N-C catalysts prepared using metalation strategies exhibit rates that bracket these values: Fe-N-C^{dm} exhibits a low HQ oxidation rate ($1.4 \times 10^{-5} \text{ mol g}^{-1} \text{ s}^{-1}$, Figure 4B, green circle), whereas Fe-N-Ctm exhibits a rate six-fold higher ($8.8 \times 10^{-5} \text{ mol g}^{-1} \text{ s}^{-1}$, Figure 4B, green diamond). The measured rates for each of these catalysts show a poor correlation with the total Fe content. Another manifestation of this observation is that the nominal turnover frequencies, defined with respect to the bulk Fe content (TOF, per total Fe), vary significantly across the full series of catalysts (Figure 4C).

The nominal TOFs in Figure 4C highlight the merits of the molecularly inspired metalation procedures outlined here.

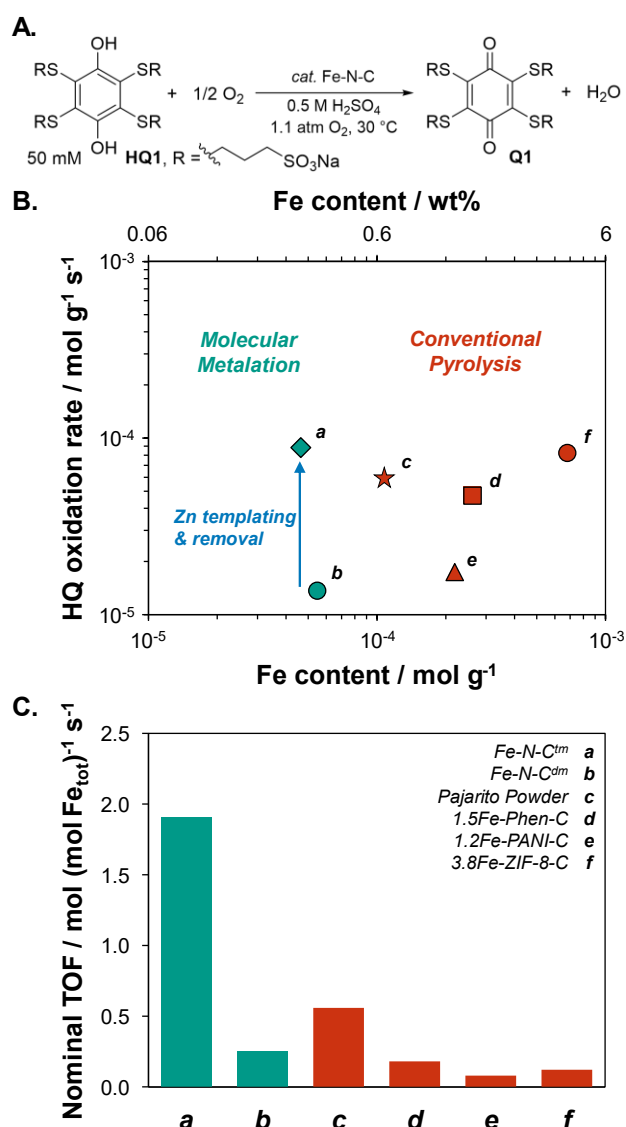


Figure 4. Kinetic assessment of the success of Fe-N-C synthesis: (A) Conditions for HQ oxidation rate measurements. (B) HQ oxidation rate (per g catalyst) as a function of bulk Fe content on Fe-N-C catalysts prepared by different routes (conventional pyrolysis, red; metalation, green). (C) Nominal TOF (per total Fe) of the same catalysts.

Although the Fe-N-C^{dm} material prepared by direct metalation (**b**) of the N-C material shows a lower mass-normalized rate than the known catalysts (**c–f**), it has a similar nominal TOF to three of them (**d–f**). The highest nominal TOF value, however, arises from Fe-N-Ctm (**a**), prepared by removal of templating Zn ions from the support prior to metalation. This catalyst exhibits a nominal TOF three-fold higher than the second-highest catalyst (**c**), reflecting a higher fraction of catalytically active Fe sites relative to total Fe in the material (e.g., FeN_x sites).⁵⁴ Similar insights are obtained from a comparison of catalytic rates (per g catalyst) for the two catalysts prepared here (Figure 4B, blue arrow). The transmetalation catalyst **a** shows much higher rate, even though the total Fe content is very similar. This outcome is again consistent with a higher density of FeN_x centers in the former catalyst.

In summary, this work demonstrates the adaptation of low-temperature molecular synthesis concepts to the metalation of nitrogen-doped carbon supports, complementing the commonly used protocols to prepare Fe-N-C catalysts. Specifically, the FeN_x sites accessed by this solution-phase metalation procedure exhibit catalytic aerobic oxidation reactivity that is similar to or exceeds the performance of Fe-N-C catalysts metalated under high-temperature pyrolysis conditions (e.g., 750–1000 °C). This work provides a foundation for numerous future opportunities, including the creation of N-C materials that have a higher density of metal binding sites, and the synthesis of M-N-C materials with metals other than Fe, including other non-precious metals as well as noble metals. Such efforts will benefit from and contribute to stronger connections between the fields of homogeneous and heterogeneous catalysis.

ASSOCIATED CONTENT

Supporting Information. Experimental procedures, characterization data, kinetic data, and supplementary discussions. This material is available free of charge via the internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interests.

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TOC Graphic

